

SWARM

BIOPOWER AI-IoT Plant Health Monitoring System

User Manual

by nanoFarm

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Dashboard monitoring, Plant HMI (process flow diagram), ESP sensor hub pairing, alerts, AI recommendations, and admin features. Production app: <https://app.swarm.co.in>

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1. Getting Started

Login

1. Open the application URL (e.g. <http://localhost:3000> or <https://app.swarm.co.in>).
2. Enter your email and password.
3. Click Sign In.

Forgot password

1. Click Forgot Password? on the login page.
2. Enter your registered email.
3. Check your email for reset instructions.

Demo credentials (Tata Steel HMI plant)

Role	Email	Password
Plant Admin	tata.admin@tatasteel.com	TataSteel@2026
Operator	tata.operator@tatasteel.com	TataSteel@2026

2. Dashboard

The main dashboard provides real-time plant monitoring:

- Plant Health Score (0-100): Green = Healthy, Yellow = Warning, Red = Critical
- Sensor cards: pH, temperature, pressure, gas flow, methane, CO₂, H₂S, NH₃
- Active alerts: latest unresolved alerts
- Gas production: current biogas output rate

Select a plant from the dropdown to switch between sites.

3. Plant HMI

The Plant HMI is the interactive process flow diagram (PFD) for the Tata Steel West Bokaro biomethanation plant. It shows equipment status, live instrument readings, animated pipeline flow, and block-level start/stop controls.

Note: Simulation mode: commands update server-side HMI state. They do not drive real PLCs or field pumps until Phase 3 field integration.

Access

Entry	Path
Sidebar	Plant HMI
Direct URL	/plant-hmi
With plant	/plant-hmi?plantId=<id>

Roles: All authenticated users can open the HMI. Plant Admin and Super Admin can send commands. Operators can view status and readings only.

Page layout

1. SWARM Core header - platform modules and simulation indicator.
2. Master output enable - energize or de-energize the plant electrical bus.
3. Process flow diagram - animated PFD with block buttons and live readings.
4. HMI page tabs - Overview, zone pages, alarms, trends, diagnostics, audit.

Master output enable (plant bus)

Action	Effect
Energize bus	Powers controllable equipment (ready, not running)
De-energize bus	Stops all equipment and disables the plant

Only Plant Admin and Super Admin see these buttons.

Block control bar

Eight equipment blocks (plus All On when available):

Button	Tag	Equipment
Belt Conveyor	BC101	Feed belt
Crusher	CH101	Crusher
Pre Treatment Tank	P-101B	Duty feed pump PW
Motor	AG101	Pre-treatment mixer / fan
Main Digester	AG102	Digester agitator
Slurry Storage Tank	P102	Slurry pump
Treatment Water Tank	P104	Treatment water pump
Equalization Tank	P103	Equalization pump

Button colors (when bus is energized)

Color	Meaning
Green	Running
Red	Off (stopped)
Yellow	Fault
Grey	Bus de-energized or idle

Click a block button to toggle START / STOP for that equipment.

Live readings panel

- Equipment - block tag, value, and run status (Off / Running / Fault).
- Instruments - FIT, PIT, LIT tags with live values when sensors are linked.

Values refresh about every 4 seconds while the HMI page is open.

Pipeline flow animation

- Animated dashed lines show water, slurry, and gas flow.
- Flow follows active blocks - pipes animate when related equipment is running.
- Fresh water feed to heater animates when the bus is energized.

PS / PW feed pumps (pre-treatment)

- PW (P-101B) - duty / working pump (Pre Treatment Tank block).
- PS (P-101A) - standby pump.

Condition	PS to PW line	PW to digester line
PW running (normal)	Off	On (when motor + digester active)
PW off, PS running	On	On
Both off	Off	Off

HMI page tabs

Tab	Purpose
Plant Overview	Process flow summary, auto-sequence, safety chain
Feed & Pretreatment	T101, BC101, CH101, T102, AG101
Digester & Feed Pumps	P-101A/B, T104, AG102
Gas System	Balloon, scrubber, generator, flare
Slurry & ETP	Slurry tanks, screw press, equalization, water
Alarm Summary	Active HMI alarms
Trends & Totals	Pressure, flow, temperature trends
Maintenance / Diagnostics	I/O quality, communication, runtime
Audit Log	Command history for the plant

Troubleshooting (HMI)

Issue	What to check
Block buttons greyed out	Energize bus first; confirm Admin role
No live readings	Sensor nodes linked; wait ~4 s poll
Flow not animating	Related block running; bus energized
Diagram not updating	Hard refresh Ctrl+Shift+R

4. ESP Sensor Hub

SWARM MODEL hub: DHT11 (temperature/humidity) + MQ5 (methane/gas).

Quick setup flow

1. Admin sets Device Unique ID in firmware and flashes once.
2. On site, connect phone to SWARM-Setup-<chipId> (password = Unique ID).
3. Open <http://192.168.4.1/setup>, enter Unique ID + site Wi-Fi, save.
4. In SWARM > Connect Device, enter ESP LAN IP + Device Password, then pair.
5. Verify readings on Dashboard.

Full guide with screenshots: [docs/ESP_SENSOR_HUB_MANUAL.md](#)

5. Plant Management (Admin)

Add a plant

1. Navigate to Plants in the sidebar.
2. Click Add Plant.
3. Fill in: Name, Type, Location, Capacity, Feedstock, Installation Date.
4. Click Save. Plants with P&ID feedstock are eligible for Plant HMI.

Plant types

- Biogas Plant
- Bio-CNG Plant
- Sanitation Plant
- STP Plant
- Organic Waste Plant
- Waste-to-Energy Plant

6. Sensor Nodes (Admin)

Register a sensor

1. Go to Sensor Nodes > Register Node.
2. Select plant, enter node name and sensor type. Save.

Monitor node health

- Battery level: should be above 20%
- Signal strength: should be above 50%
- Status: Active, Inactive, Faulty, or Offline

7. Analytics

1. Select plant and sensor type.
2. Choose time range: Hour, Day, Week, Month.
3. Interactive chart displays trend data.

8. Alerts

Alert severities

- Critical: immediate action (red)
- Warning: attention needed (yellow)
- Information: for awareness (blue)

Managing alerts

1. View by status: Active, Acknowledged, Resolved.
2. Click Acknowledge to mark as seen.
3. Click Resolve when issue is fixed.

9. AI Recommendations

- Health score (0-100)
- Issue detection: acidification, overfeeding, gas yield reduction
- Actionable recommendations for operators

Example: "Reduce feedstock input by 15%. Increase mixing cycle frequency."

10. Predictive Maintenance

- Remaining Useful Life (days)
- Estimated Failure Date
- Equipment Health Percentage

Equipment types: Pump, Blower, Agitator, Compressor, Sensor

11. Reports

1. Click Generate Report.
2. Select plant, report type, and format (PDF/Excel/CSV).
3. Download from the reports list.

12. User Roles

Feature	Super Admin	Plant Admin	Operator
View Dashboard	Yes	Yes	Yes
View Plant HMI	Yes	Yes	Yes
Control Plant HMI	Yes	Yes	No
Manage Plants	Yes	Yes	No
Manage Sensors	Yes	Yes	No
Connect Device (ESP)	Yes	Yes	Yes
View Alerts	Yes	Yes	Yes
AI Recommendations	Yes	Yes	Yes
Generate Reports	Yes	Yes	Yes
Manage Users	Yes	No	No
System Settings	Yes	No	No

13. Notifications

The bell icon in the navbar shows active alert count. Click to open the Notification Center.

14. Settings (Super Admin)

- Alert thresholds: pH, temperature, pressure limits
- MQTT broker: IoT message broker settings
- Email server: SMTP for notifications
- AI settings: analysis interval
- Report schedule: automated report timing

15. Android APK (phone and tablet)

- Debug APK: releases/SWARM-debug.apk
- Build guide: docs/ANDROID_APK.md

Install by copying APK to device. Backend URL from VITE_API_URL in frontend/.env.production.

16. Development - Run Locally

Frontend

```
cd frontend
npm install
npm run dev
URL: http://localhost:3000
```

Backend

```
cd backend
./mvnw.cmd spring-boot:run -Dspring-boot.run.profiles=dev
API: http://localhost:8080/api
```

- Plant HMI: <http://localhost:3000/plant-hmi>
- Git push: `git push swarm main`
- LAN access: `http://<your-PC-IP>:3000/plant-hmi`

End of SWARM User Manual